Microscopy Morphology Under Domain Shift

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Purpose and intended demonstration

Measure how a neural morphology classifier changes across synthetic acquisition conditions. This is a computational biology learning project with synthetic or fictional inputs, not a clinical or experimentally validated system.

How the implementation works

Use hand-engineered intensity histogram and local-gradient summaries as inputs to a two-layer tanh classifier. Train on bright imaging domains and compare raw versus per-image normalized features on a shifted domain.

Evaluation and evidence

Split entire acquisition domains. Report held-out-domain accuracy for both preprocessing choices with a majority baseline, avoiding pixel-level leakage.

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Setup and commands

Requires Python 3.10 or newer and NumPy. Create a virtual environment: python -m venv .venv. On Windows use .venv\Scripts\python.exe if activation is unavailable. Install with python -m pip install -r requirements.txt. Run python app.py, then open results/report.html in your browser. Run python -m unittest discover -v to execute tests. The default seed is 42; change it with --seed. Replace sample_input.json using --input path-to-file.json, preserving its schema. Algorithm-only projects generate synthetic arrays from the seed rather than consuming external files.

Architecture and file responsibilities

CLI arguments -> JSON input or seeded generator -> domain pipeline -> validation or evaluation -> JSON and HTML artifacts. Imported functions can be used in a notebook or a separate application. Model checkpoints preserve learned arrays where applicable; see app.py for parameter order. This package does not include a production web service. Each project is independent and includes its own helper code so the ZIP can be extracted and run alone. app.py holds the experiment and domain functions: generate, features, fit, predict, run. kit.py contains reporting, input loading and only the retrieval or generation helpers needed by this project. sample_input.json contains editable inputs. test_app.py checks domain behaviors; test_reporting.py checks metrics and HTML output boundaries. results/metrics.json records outputs and results/report.html provides a browser-readable report. Numeric projects also save arrays or learned parameters to results/artifacts.npz; inspect its keys with numpy.load. MODEL_CARD.md explains intended use and limits. .github/workflows/tests.yml runs tests and the fixture experiment. No API key is needed.

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Limitations and next improvements

Labels are fictional sparse versus dense patterns, not diseases or diagnoses. Normalization can erase genuine biology and must be assessed on real data. Use licensed morphology data, specimen-group splits, stain or instrument perturbations, calibration and domain-adaptation baselines.

Interview walkthrough

Present the problem and demonstrate one normal example, then one failure or boundary case. Explain the baseline or fixture mode, the split or validation design, and one limitation before discussing an extension. Be ready to trace data through the functions listed above. Explain why the chosen metric matches the task and why the available results cannot establish production performance. Add your own documented experiment and be precise about what you implemented or changed. These are AI-assisted educational projects; filenames and synthetic scores are not evidence of client work.

Research context and reading

These topics were selected from research directions reviewed on 22 October 2026. Recent activity suggests useful skills to develop; it is not a forecast of hiring, commercial success or scientific validity. The implementation is a small educational baseline and does not reproduce or claim the performance of any cited foundation model. Reading: CellSAM: a foundation model for cell segmentation (2025): https://www.nature.com/articles/s41592-025-02879-w.

Verified build results

project

"Microscopy Morphology Under Domain Shift"

domain results

{"raw": {"accuracy": 0.535, "precision": 0.535, "recall": 1.0, "f1": 0.6970684039087948, "tp": 107, "fp": 93, "fn": 0, "tn": 0}, "normalized": {"accuracy": 1.0, "precision": 1.0, "recall": 1.0, "f1": 1.0, "tp": 107, "fp": 0, "fn": 0, "tn": 93}}

majority baseline accuracy

0.465

training brightness

1.0

test brightness

1.8

labels

"Fictional sparse/dense morphology; no diagnostic labels"

See results/metrics.json for all observations and TEST_RESULTS.txt for the executed tests. All datasets and replay responses shipped here are synthetic. Results can vary across seeds and numerical library versions.